

Malek Aziz Hassayoun

AI Engineer | Cybersecurity & Offensive Automation | LLM Systems

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Software Engineering professional with hands-on experience in **AI systems**, **LLM architectures**, **cybersecurity**, and **software architecture and system design**. Delivered real-world projects across **8+ internships and research experiences**, covering **High-Level Design (HLD)**, **Low-Level Design (LLD)**, microservices architectures, AI-driven security automation, blockchain systems, smart-grid security, and cloud core network infrastructure. Currently contributing to **Huawei's 5G Core (5GC) Expansion**. Ranked in the **top 6% on TryHackMe**, active IEEE project contributor, and open-source Project. Passionate about **scalable architectures**, XAI, federated learning, and AI-powered security systems.

PROFESSIONAL EXPERIENCE

Cloud Core Network Intern **Apr 2026 – Present**
Huawei  *Lac 2 / Tunisia*

- Contributed to the **expansion and deployment of Huawei's 5G Core (5GC) network infrastructure**, supporting integration, configuration, and network rollout activities
- Provided **technical support and operational assistance** for cloud core network ensuring service continuity and efficient network operations

Professional AI Instructor (Part-time) **Jan 2026 – Present**
TCodeX (TeachCode X)  *Remote*

- Instructed **100+ students** across LLMs & RAG, Prompt Engineering, AI Agents, and AI-driven cybersecurity, sustaining a **90%+ completion rate**
- Produced a **72-hour full-pack curriculum** across **8 modules**, cutting onboarding time by **30%**; validated through personal **NVIDIA certification** completion in Deep Learning, RAG Agents, NLP, and Anomaly Detection
- Supervised **12+ capstone projects**; **4 shipped** as open-source tools on GitHub

Freelance Security & Architecture Consultant **Jan 2026 – Present**
Self-Employed  *Remote*

- Conducted security audits for **3+ clients**, uncovering an average of **8 critical vulnerabilities** per engagement with full remediation roadmaps
- Planned secure cloud architectures improving infrastructure resilience scores by **35%** on average

AI-Powered Web Pentesting Intern **Jun – Aug 2025**
OXYNAS  *Paris / Remote*

- Engineered an AI pentesting agent that **cut manual testing time by 60%** and flagged **85% of OWASP Top-10 threats** autonomously
- Systematised the vulnerability-to-report pipeline, compressing report generation from **4 hours to under 45 minutes**

Blockchain & AI Research Intern **Jul – Sep 2025**
ESPRIT Research Lab  *Tunis, Tunisia*

- Developed a decentralised smart grid security system; realised **49.5 ms benchmark latency** for AI anomaly detection, **32% below target**
- Authored internal research report adopted as reference architecture for two follow-on lab projects

WiFi / Bluetooth Pentesting Intern **Jun – Aug 2024**
ESPRIT  *Tunis, Tunisia*

- Executed **15+ wireless attack scenarios** (WPA2 cracking, BLE sniffing, deauth floods); documented findings in a structured threat model
- Deployed countermeasures that **reduced the wireless attack surface by 40%**, confirmed through independent re-testing

Web Application Pentesting Intern **Jun – Sep 2024**
TALAN Tunisia  *Tunis, Tunisia*

- Mechanised OWASP ZAP scans with custom Python scripts across **10+ web targets**, lifting cycle efficiency by **40%**
- Uncovered **23 actionable vulnerabilities** — including **3 high-severity SQLi** — and delivered remediation reports to dev teams

Junior Career Center Assistant (Volunteer) **Jan 2024 – Dec 2025**
ESPRIT Career Center  *Tunisia · On-site*

- Co-organised **6 career events** (job fairs, company forums, mock interviews) reaching **500+ student attendees**
- Reviewed CVs and coached **100+ students** on professional positioning, contributing to improved internship placement rates

SELECTED PROJECTS

Cognicare – AI Digital Health  **Dec 2025**
Next.js 14 · FastAPI · Phi-4 Multimodal · OpenCV IEEE / Open-source

- Attained **92% diagnostic accuracy**; lifted user engagement by **35%** via real-time gamified therapy tracking

ISO 9001 ERP – Quality Automation **Oct 2024 – May 2025**
SpringBoot · Angular 19 · PostgreSQL · Docker · K8s Cocosunult

- Slashed documentation overhead by **60%**; compressed audit-preparation cycles **5x** via end-to-end process mechanisation

Cyber Sentinel – Security Platform  **2024**
Python · Semgrep · Nmap · AI/ML Personal

- Unified code scanning, network recon, and AI vulnerability scoring; reduced manual triage time by **50%**

TECHNICAL SKILLS

AI & ML: Phi-4, Llama 3.1, RAG, PyTorch, Transformers, CV, XAI, Federated Learning, LLMops, n8n

Cybersecurity: Pentesting, OWASP ZAP, Metasploit, Nmap, Burp Suite, Snort, Suricata, Wireshark, DevSecOps

Architecture: HLD, LLD, System Design, Microservices, Distributed Systems, Design Patterns, REST APIs, Scalability

Algorithms: Data Structures, Algorithm Design, Graph Algorithms, Optimization, Complexity Analysis, Consensus Algorithms

Blockchain: Hedera Hashgraph, HBAR, HCS, HTS, Smart Contracts, Solidity, EVM, Web3, DLT, PoA, PoF, BFT

Development: Python, Java, Spring Boot, Angular 19, React, Next.js, Node.js, Docker, Kubernetes

Data & Research: EDA, Model Benchmarking, Literature Review, Experimental Evaluation, Statistical Analysis

CERTIFICATIONS

Hedera / Blockchain: HBF *Apr 2026* · HCF *Apr 2026* · Hashgraph Developer *May 2026* · Blockchain Fundamentals

NVIDIA: Deep Learning · RAG Agents · Custom LLMs · Anomaly Detection · NLP · Diffusion · Predictive AI · Jetson Nano

Cisco: Ethical Hacker · Cybersecurity Essentials · Junior Cyber Analyst · Networking Basics

Others: Oracle AI Vector · AWS Cloud · GitHub Foundation · AppSec (CAP) · Scrum · IBM Data Viz · Python 1

EDUCATION

Ingénieur Génie Logiciel — Master Level **2023 – Present**
ESPRIT — École Supérieure Privée d'Ingénierie Ariana, Tunisia

- Advanced AI & LLM Systems, Cloud-Native DevSecOps, Cybersecurity

Global AI Education Program **Sep – Nov 2025**
Samsung Innovation Campus Online

- ML Foundations, Statistics, Mathematics for AI

Preparatory Studies for Engineering **2020 – 2023**
Faculty of Sciences, Monastir Tunisia

LANGUAGES

Arabic: Native | **English:** B2 Professional | **French:** B2 Professional